5% **myoglobin,** various **heme and non heme enzymes** (5%). The remaining iron is found in body storage as **ferritin** (20%) and **hemosiderin** (10%) the two major iron storage proteins.

In developing countries estimated prevalence of anemia is 39 percent in children and 42 percent in women. In Asia 12.8 of maternal mortality deaths are caused by anemia.

- **Forms of dietary iron**

  Heme iron: Heme iron is the iron associated to the protein globin to form hemoglobin and is found in flesh foods only

  Non heme iron: This form is present in all plant sources in addition to 60% of animal sources.

Heme absorption varies from 15-35 % depending on the iron status of the consumer. Non heme iron absorption can vary widely from less than 1% to more than 90% but usually in the range of 1-20%.

- **Absorption:** Several factors favor and inhibit iron absorption

  Factors favoring absorption of Iron:
  - Body needs,
  - Ascorbic acid,
  - Animal tissues,
  - Pregnancy,
  - Low iron status,
  - Low heam iron intake.

Factors decreasing the absorption of iron:
- Binding agents like fiber, phosphates, phytates and oxalates,
- High calcium intake,
- Achlorohydria (low gastric acid)
- Infection
- Gastrointestinal disease.

- **Functions:** Iron has varied significant functions in human body. They include:
  - Transport and storage of oxygen where each gram of hemoglobin contains about 3.34 mg of iron.
  - As myoglobin iron is required for oxygen storage in muscle.
  - Iron acts as a cofactor of enzymes.
  - It is a component of cell enzyme systems that oxidize glucose and other energy yielding nutrients.
  - Production of immune cells that attack foreign bacteria invading the body.
  - Positive iron balance is necessary for continued growth.
  - To build reserves for physiologic stress during adolescence for both boys and girls.
  - Necessary for brain development, cognitive function, the synthesis and breakdown of neurotransmitters.

- **Food sources:**

  Rich sources of iron are cereals, millets, pulses and green leafy vegetables. Of the cereal grains and millets bajra and ragi are very good sources of iron. Other sources of plant foods include manathakali

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